Calculus & Optimization

Formula sheet + conditions

A formula is useful only with its assumptions. This sheet pairs each relationship with a reminder of a limitation or common mistake. Read the linked lesson notes for derivations and examples. Symbols: A^{T} is transpose, A^{-1} inverse, A^{+} pseudoinverse, $||\cdot||_2$ Euclidean/operator 2-norm, I identity, λ eigenvalue, σ singular value. Derivatives use radians and natural logarithms unless stated otherwise.

Derivative table

$(x^r)' = r\,x^{r-1}$; $(c)' = 0$

$(e^x)' = e^x$; $(a^x)' = a^x \ln a$

$(\ln|x|)' = 1/x$; $(\log_a x)' = 1/(x \ln a)$

$(\sin x)' = \cos x$; $(\cos x)' = -\sin x$

$(\tan x)' = \sec^2 x$; $(\cot x)' = -\csc^2 x$

$(\sec x)' = \sec x \tan x$; $(\csc x)' = -\csc x \cot x$

$(\arcsin x)' = 1/\sqrt{(1-x^2)}$; $(\arccos x)' = -1/\sqrt{(1-x^2)}$

$(\arctan x)' = 1/(1+x^2)$

Use radians. For exponent/log base a require $a>0, a\neq 1$. Respect each real domain; $\ln|x|$ excludes zero, inverse sine/cosine formulas require $|x|<1$, and denominators must be nonzero.

Rules that combine derivatives

$(au+bv)' = au' + bv'$

$(uv)' = u'v + uv'$

$(u/v)' = (u'v - uv')/v^2$

$(f(g(x)))' = f'(g(x))g'(x)$

$(f^{-1})'(y) = 1/f'(x)$, $y = f(x)$

$(u^v)' = u^v(v' \ln u + v\,u'/u)$

a,b are constants. Quotient rule needs $v\neq 0$; inverse derivative needs a locally differentiable inverse and $f'\neq 0$; the logarithmic power formula assumes $u>0$.

Limits worth recalling

$\sin x/x \to 1$; $\tan x/x \to 1$ as $x \to 0$

$(1-\cos x)/x^2 \to 1/2$

$(e^x-1)/x \to 1$; $\ln(1+x)/x \to 1$

$(1+x)^{1/x} \to e$

$\lim f/g = \lim f'/g'$ under L'Hôpital hypotheses

The first four lines concern $x \to 0$. L'Hôpital requires an appropriate 0/0 or ∞/∞ form, differentiability near the point, nonzero denominator derivative and an existing derivative-ratio limit; it is not a universal quotient rule for limits.

Taylor and standard series

$T_n(x)=\Sigma_{k=0}^{n} f^{(k)}(a)(x-a)^k/k!$

$|Rn|\le M|x-a|^{n+1}/(n+1)!$ if $|f^{(n+1)}|\le M$ between a and x

$e^x=1+x+x^2/2!+x^3/3!+\ldots$

$\sin x=x-x^3/3!+x^5/5!-\ldots$

$\cos x=1-x^2/2!+x^4/4!-\ldots$

$\ln(1+x)=x-x^2/2+x^3/3-\ldots$

$1/(1-x)=1+x+x^2+\ldots$

$(1+x)^\alpha=1+\alpha x+\alpha(\alpha-1)x^2/2!+\ldots$

Exponential, sine and cosine series hold for all real x. Logarithm: $-1<x\le 1$. Geometric: $|x|<1$. General non-terminating binomial series: $|x|<1$, with endpoints checked separately.

Extrema and iteration

Interior differentiable extremum $\Rightarrow f'=0$

$f'=0, f''>0 \Rightarrow$ strict local minimum

$f'=0, f''<0 \Rightarrow$ strict local maximum

GD: $x_{next}=x-\eta f'(x)$

Newton optimization: $x_{next}=x-f'(x)/f''(x)$

$f=\frac{1}{2}ax^2, a>0$: GD convergence if $0<\eta<2/a$

$f''=0$ is inconclusive. For global extrema on a closed bounded interval, compare endpoints, stationary points and nondifferentiable points. Newton needs nonzero curvature and does not guarantee a minimum.

Supporting bridge: integrals and gradients

$\int x^p dx=x^{p+1}/(p+1)+C,\ p\ne -1$

$\int dx/x=\ln|x|+C;\ \int e^x dx=e^x+C$

$\int \sin x\,dx=-\cos x+C;\ \int \cos x\,dx=\sin x+C$

$\int_a^b f=F(b)-F(a)$ if $F'=f$

$\int u\,dv=uv-\int v\,du$

$\nabla(\frac{1}{2}x^TAx-b^Tx)=Ax-b$ for $A=A^T$

$\nabla\frac{1}{2}||Mx-y||^2=M^T(Mx-y)$

These are supporting topics for probability and ML. Antiderivatives apply on valid real-domain intervals. The definite integral is signed accumulation; a general non-symmetric quadratic uses its symmetric part.

Lesson-by-lesson recall

Functions & algebra refresh

Domain of $\sqrt{(x-1)}/(x-3)$: $[1,3) \cup (3,\infty)$

A simplified expression can have a larger domain than the original function.

$(f \circ g)(x)=f(g(x)) \cdot f^{-1}(f(x))=x$ on an invertible domain

$f^{-1}(x)$ denotes inverse function, not $1/f(x)$.

$\ln(ab)=\ln a+\ln b \cdot a^x=e^{x \ln a} \cdot 180°=\pi$ radians

The standard sin and cos derivatives assume radians, and logarithm rules require positive real arguments.

Limits & continuity

$\lim_{x \to a} f(x)=L \Leftrightarrow f(x)$ approaches L from both sides of a

$f(a)$, $\lim_{x \to a} f(x)$, and the existence of a limit are three separate questions.

$\sin x/x \to 1 \cdot \ln(1+x)/x \to 1 \cdot (e^x-1)/x \to 1 \cdot (1+x)^{1/x} \to e$

0/0 is a signal to investigate; it is not zero and it is not automatically undefined as a limit.

Continuous at a $\Leftrightarrow \lim_{x \to a} f(x)=f(a) \cdot$ continuous on $[a,b] \Rightarrow$ extrema attained

A continuous function may have corners, and a root guaranteed by a sign change need not be unique.

Differentiation from scratch

$f'(a)=\lim_{h \to 0} [f(a+h)-f(a)]/h \cdot$ tangent: $y=f(a)+f'(a)(x-a)$

The derivative is a limit, not the quotient with $h=0$ substituted directly.

$(uv)'=u'v+uv' \cdot (u/v)'=(u'v-uv')/v^2 \cdot (f \circ g)'=(f' \circ g)g'$

The derivative of a product is not the product of derivatives. Do not omit the inner derivative.

$y=u^v>0 \Rightarrow y'=u^v(v' \ln u+v\, u'/u) \cdot f^{(n)}$ = n-th derivative

Applying the ordinary power rule to x^x treats the changing exponent as constant and gives the wrong result.

What derivatives reveal

$\exists c \in (a,b)$: $f'(c)=[f(b)-f(a)]/(b-a)$

Rolle's theorem does not apply to $|x|$ on $[-1,1]$ because differentiability fails at zero.

$f'>0 \Rightarrow$ increasing $\cdot f''>0 \Rightarrow$ convex $\cdot$ inflection requires concavity change

A zero second derivative alone does not prove an inflection or classify a stationary point.

$\Delta f \approx f'(a)\Delta x \cdot |f(x)-L(x)| \leq M|x-a|^2/2$ when $|f''| \leq M$

An approximation near a point can be poor far away, even if the derivative formula is exact.

Taylor series & approximation

$T_n(x)=\Sigma_{k=0}^{n} f^{(k)}(a)(x-a)^k/k!$

Do not omit factorial denominators or confuse polynomial degree with the number of nonzero terms.

$|Rn(x)|\le M|x-a|^{n+1}/(n+1)!$ · $1/(1-x)=\Sigma x^k$ for $|x|<1$

A series identity has a domain of convergence; plugging in a value outside it is invalid.

$f(a+h)-f(a)\approx f^{(n)}(a)h^n/n!$ for the first nonzero derivative order n

When a numerator's leading terms cancel, a lower-order approximation can erase the very term you need.

Maxima, minima & optimization

Global search on [a,b]: endpoints + interior $f'=0$ + interior nondifferentiable points

Solving $f'=0$ alone misses corners and boundary optima, and includes non-extrema.

f convex and $f'(x^*)=0 \Rightarrow x^*$ global minimum · strict convexity ⇒ at most one minimizer

A formula without its feasible domain is an incomplete optimization model.

GD: $x_{next}=x-\eta f'(x)$ · Newton: $x_{next}=x-f'(x)/f''(x)$

Newton's root method $x-f/f'$ solves $f=0$; Newton's optimization method $x-f'/f''$ seeks stationarity. They are different targets.

Supporting bridge: integration & gradients

$\int_a^b f=F(b)-F(a)$ · $\int u\,dv=uv-\int v\,du$ · $\int 2xe^{x^2}dx=e^{x^2}+C$

An indefinite integral needs +C; a definite integral is a number with signed area, not necessarily positive.

$\nabla f=(\partial f/\partial x_i)$ · $Duf=\nabla f^T u$ $(||u||=1)$ · $H_{ij}=\partial^2 f/(\partial x_i \partial x_j)$

Positive semidefinite Hessian at a single stationary point does not alone prove a minimum for a general nonlinear function.

$\beta^*=\text{argmin}\ \frac{1}{2}||X\beta-y||^2$ · $X^Tr=0$ · $H=X^TX$

A prediction residual can be nonzero at the optimum; what vanishes is its projection onto every design-column direction.